

Techno-economic analysis to develop the first wind farm in the Egyptian western desert at Elkharga Oasis

Mohamed A. Abdelrahman,*^{ORCID} Radwan H. Abdel-Hamid, Maged Ahmed Abo Adma and Mohamed Daowd

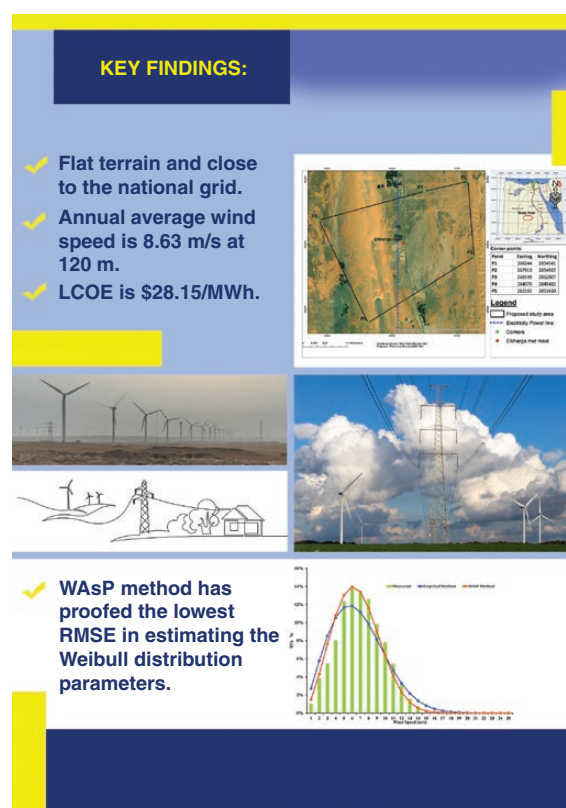
Electrical Power and Machines Department, Faculty of Engineering, Helwan University, Cairo, Egypt

*Corresponding author. E-mail: m.nrea2010@yahoo.com

Abstract

During the past few years, Egypt has increased the share of electricity generation from renewable energy sources to achieve the renewable-energy strategy. The Egyptian Ministry of Electricity and Renewable Energy has set an ambitious target to increase the share of renewable energy among the total energy mix to be 20% by 2022 and 42% by 2035. This target will be achieved using several approved policies such as feed-in tariffs, build-own-operate and independent power producers under a merchant scheme. This paper presents a full analysis of wind-potential characteristics at Elkharga Oasis in Egypt based on an actual wind-measurement campaign taken by a met mast at two height levels of 10 and 25 m, respectively. The measurements show an average annual wind speed of 5.72 m/s at 10-m height and 6.53 m/s at >25-m height. The estimated wind speed, however, is 8.38 m/s at 120 m, which is technically convenient to generate electricity from wind energy. In this paper, the wind potential in the area is assessed using the statistical method of the Weibull probability density function. The different methods to estimate the Weibull distribution parameters are presented and the relevant accuracy is verified based on the root mean square error. A techno-economic assessment and selection of the optimal wind-turbine size with ideal micro-sitings are performed using the software package windPRO. Different cases of study for three typical wind-turbine models with various rotor diameters, power capacity, layout configuration and annual yield are assessed at the site. The selection of the most convenient wind turbine is determined based on the IEC 61400 standard criteria and the turbine that produces the lowest levelized cost of electricity.

Graphical Abstract



Keywords: wind energy; wind-resource assessment; Weibull distribution; windPRO; levelized cost of electricity

Received: 4 November 2021. Accepted: 12 January 2022

© The Author(s) 2022. Published by Oxford University Press on behalf of National Institute of Clean-and-Low-Carbon Energy

This is an Open Access article distributed under the terms of the Creative Commons Attribution-NonCommercial License (<https://creativecommons.org/licenses/by-nc/4.0/>), which permits non-commercial re-use, distribution, and reproduction in any medium, provided the original work is properly cited. For commercial re-use, please contact journals.permissions@oup.com

Introduction

The annual energy output of a wind farm is subject to several factors such as site wind potential, number of turbines, turbine spacing and loss factors including wake losses and electrical and environmental losses. The siting of wind turbines should carefully consider positioning the turbines at the highest production spots with the lowest wake interaction and complying with all environmental restrictions. The wake-effect phenomenon reduces the wind speed downstream of each wind turbine causing a reduction in the output power of the downstream turbines. The proper design of the wind farm should verify different alternatives of turbine sizes and turbine distributions. The final selection of the most convenient wind turbine should be according to techno-economic iterations. The case study of this research is proposing a wind farm at Elkharga Oasis located in the Egyptian western desert. The proposed site is an area of 19 km² with dimensions of 4.3 km in the west direction, 5.8 km in the north direction, 2.8 km in the east direction and 4.8 km in the south direction. The coordinates of the corners of the proposed site are shown in Fig. 1. The research includes an optimal selection of the most convenient wind turbines for a proposed wind farm of 50 MW using three wind-turbine models. In addition, two alternatives of wind-turbine spacing have been properly tested for each turbine model to select the most appropriate turbine layout that leads to the highest yield and lowest levelized cost of electricity.

The Egyptian government, in cooperation with Risø National Laboratory, Roskilde, Denmark, have studied the wind resource in Egypt and issued the Egyptian wind atlas in 2006. This atlas is considered one of the most complete numerical wind atlases ever known, which allows wind-resource assessment and siting anywhere in Egypt. The wind-resource maps cover the whole area of Egypt including mountains and remote desert. The resource maps have been calculated based on two methods. The first was done using the traditional wind-atlas method based on actual measurements from >30 measurement stations distributed over Egypt. The second one is a numerical wind atlas based on long-term data and a mesoscale model [1].

Shields et al. [2] have estimated the impact on the levelized cost of energy (LCOE) of turbine rating ranges from 6 to 20 MW and total installed capacities of between 250 and 2500 MW using techno-economic costs along with the expected annual yield. These results show that upsizing the turbine to 20 MW with total installed capacity of 2500 MW will lead to a reduction in the levelized cost of electricity of 23% in comparison with the international average of wind-turbine prices installed in 2019. The main drivers of LCOE are the decreases in the quantities of the balance of plant, operation and maintenance (O&M) costs, optimized export systems and novel O&M strategies.

Alemzero et al. [3] have studied wind-energy development in Africa including policy incentives and obstacles. The development growth of wind projects has rapidly increased from 2009 to 2018 by 740%. The main drivers of that growth are technology maturity, cost reduction and political incentives. The average levelized cost of electricity has reached 7 US cent/kWh in 2018 drivers; however, it was previously 10 US cent/kWh in 2010.

Mohamadi et al. [4] have assessed the wind resources in 22 areas in east Iran. The wind assessment has included estimation of Weibull parameters, power density and mean wind speed. The monthly variation in electricity demand matches the fluctuation of wind speed. The estimation of the expected annual yield and economics of all sites have been done using four commercial wind-turbine models. The outcome of the economic analysis

shows profitability of investing in wind projects in the eastern part of Iran.

Dabar et al. [5] have presented the wind potential of Djibouti by analysing measured wind data of eight masts in Djibouti and three sites indicate promising wind potential. Long-term correlation using online data has confirmed the seasonal stability of wind speed over the years. The technical feasibility of the selected three wind farms with a total capacity of 275 MW has been done using the windPRO program and two wind turbines matching the IEC61400-1 standard. The economic assessment was performed based on the present value cost method and the electricity cost, which typically varied from 7.03 to 9.67 US cent/kWh among sites.

Himri et al. [6] have used WASP (Wind Atlas Analysis and Application Program) and RETScreen tools to properly assess the economic viability of establishing wind farms in south-west Algeria. The measured wind data collected by the Algerian Meteorological Office have been used to identify the wind characteristics such as mean wind speed, dominant wind direction, frequency distribution and the Weibull distribution parameters (scale and shape factors). The good wind potential of the Adrar site produces a capacity factor of 36% which leads to viable project economics with a cost of electricity of 3.25 US cent/kWh and the simple payback period is 3.9 years.

Bilal et al. [7] have assessed the wind potential and energy yield in north-west Africa, focusing on analysing daily and seasonal wind potential. The wind data are measured at 10-minute intervals with three different height levels over 1 year and collected from eight locations. Turbulence intensity, power density, power-law exponents and Weibull parameters were calculated based on the monthly and daily wind analyses. Results show that the power-law exponent is higher where the turbulence index is low. According to techno-economic analysis findings, the most convenient wind turbines for all sites is Ecotecnia-48, which leads to a lower cost of electricity, varying from 0.0187 to 0.0596 Euro/kWh.

El Khchine et al. [8] have properly evaluated the wind-energy potential in Taza and Dakhla cities, Morocco using statistical methods of the Weibull probability density function. The Weibull distribution parameters (scale and shape factors) have been estimated using various methods such as wind variability, standard deviation, power density and the WASP tool. The WASP and wind-variability methods give better fittings of the Weibull shape and scale factors.

Yllmaza et al. [9] have properly investigated the capability to generate electricity from wind energy in Turkey to meet energy demand. The study motivates the RE power generation of a specific location and improves the awareness of the economic advantage from wind projects. The wind data have been collected by the General Directorate of State Meteorology in Turkey and assessed using the WASP simulation program.

Katinas et al. [10] have studied wind characteristics and power density at different locations in Lithuania using 2 years of measured data. The Weibull scale and shape factors were estimated using four methods and the wind characteristics and power density were calculated accordingly. Some methods have demonstrated poor accuracy at certain geographical areas with different wind speeds. It was found that most of the calculation methods based on probability density functions give a good reliable result. In order to evaluate the goodness of Weibull parameters among various methods, mean square error, chi-square test, relative error and coefficient of determination were accurately estimated.

Rogers et al. [11] have gently explained that the electricity prices at all small island developing states are often high whilst

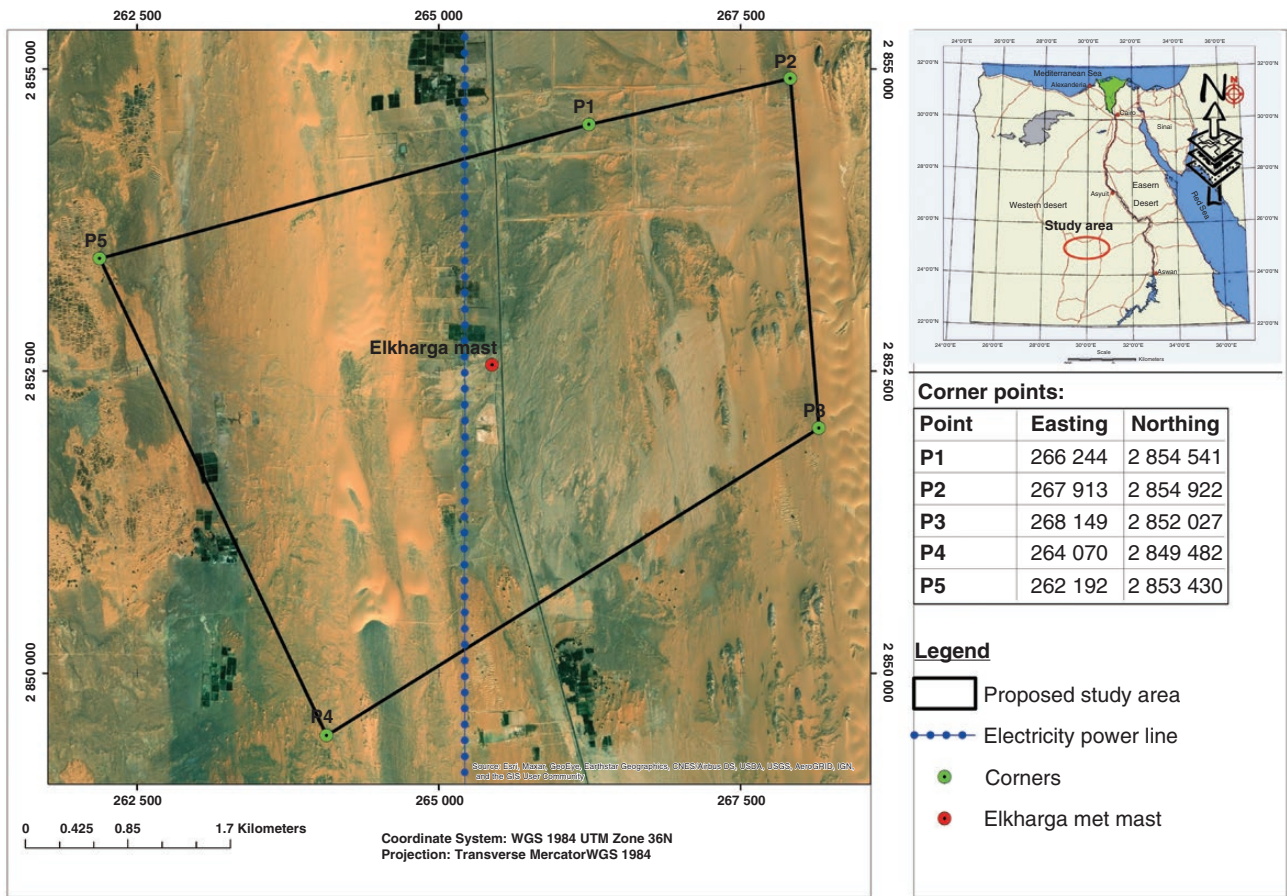


Fig. 1: Location of the proposed study area and the existing infrastructures [1].

being granted with abundant wind potential. Wind energy in common is the most mature technology among other modern renewable-energy technologies. It is recommended to keep historical hourly weather data available in order to promote comprehensive studies for the island. The research has addressed the key barriers to developing wind projects at the islands in comparison with mainland countries.

Elia *et al.* [12] have studied the cost evolution of wind-energy technologies that has rapidly dropped during the last 20 years. Many remarkable drivers leading to this reduction in cost have been precisely defined such as reduction in materials, labour cost, legal framework and financial expenditures that led to a 30% reduction in wind-turbine prices from 2005 to 2017. Policy incentives such as appropriate regulations and supporting schemes are very important for growing sustainably of the wind industry.

In Lithuania, Katinas *et al.* [13] have reviewed the current wind situation of electricity generation from wind energy and the future strategy. The wind-resources map of Lithuania has been developed using 10 years of measurement data. In addition, 19 measurement stations with proper long-term data correlation have been analysed and the results show that the strip area near the Baltic Sea is more convenient to develop wind power projects. The best site selection was done by estimating the capacity factor of the various regions and choosing the area with the highest capacity factor.

In Egypt, Hamouda [14] has examined the economic feasibility of installing a stand-alone single wind turbine of type NEG Micon60/1000 by an independent power producer. The loca-

tion of the proposed site is close to Cairo International Airport located 22 km north-east of Cairo centre. The mean wind speed is 3.43 m/s at 10 m above ground level. However, financial analysis of the humble wind resource at the site has typically shown that under certain conditions such as increasing electricity pricing by 5% and with the discount rate equal to or below 8%, the project can achieve financial viability.

The International Electrotechnical Commission standard IEC 61400 has outlined the minimum design necessities for wind turbines. The external conditions that should be considered in wind-turbine generator (WTG) design are dependent on the intended site or site type for its installation. The wind-turbine classes are categorized based on two parameters, which are the annual average wind speed and the turbulence intensity at a wind speed of 15 m/s, as shown in Table 1 [15, 16] where the values apply at hub height and v_{av} (m/s) represents the annual average wind speed at the turbine hub height, A refers to a site category of higher turbulence intensity, B refers to a site category of lower turbulence intensity and I_{15} is the turbulence intensity at a wind speed of 15 m/s.

Table 1: Basic parameters for WTG classes

WTG class	Class I	Class II	Class III	Class IV
v_{av} (m/s)	10	8.5	7.5	6
A I_{15} (-)	0.18	0.18	0.18	0.18
B I_{15} (-)	0.16	0.16	0.16	0.16

1 Research methodology

The specific methodology of the research is based on techno-economic analysis of the wind farm by assessing the measured wind data based on the Wind Atlas of Egypt and using statistical methods of the Weibull probability density function through various methods. This will lead to verification of the most accurate method among them to classify the site according to the IEC 61400 standard and selection of a shortlist of turbines matching the standard. After selecting the turbine models, we accurately estimated the annual expected yield from the wind farm and calculate the levelized cost of electricity from each turbine that will be the economic basis for selecting the most convenient turbine for the site. The specific sequence of research methodology can be summarized as following:

- developing methods for estimating Weibull distribution parameters;
- proper verification and comparison between the different methods;
- extrapolations of Weibull parameters from met-mast height to the wind-turbine hub height;
- comprehensive assessment of wind characteristics at the study area;
- wind-turbine selection;
- the proposed wind-farm design approach;
- calculation methods of the expected annual energy production (AEP) of the proposed Elkharga wind farm;
- wind-farm design methodology and preferred methods for accurately estimating the annual production;
- economic investigation of the Elkharga wind turbines based on levelized cost of electricity.

1.1 Developing methods of estimating Weibull distribution parameters

Many statistical distributions have been used to analyse and describe wind resources, such as Weibull and Rayleigh distributions. Those statistical distributions have demonstrated a sufficient accuracy in fitting the measured wind-speed data and prevailing wind-speed characteristics; accordingly, the wind potential has been assessed in this research using the Weibull probability density function and cumulative Weibull distribution function, as shown in Equations (1) and (2), respectively [17–19]:

$$f(v) = \left(\frac{k}{C}\right) \left(\frac{v}{C}\right)^{k-1} \exp\left(-\left(\frac{v}{C}\right)^k\right) \quad (1)$$

$$F(v) = 1 - \exp\left[-\left(\frac{v}{C}\right)^k\right] \quad (2)$$

where k and C are the Weibull shape and scale parameters, respectively, and v is the wind speed (m/s).

In most wind conditions, the k -factor varies from 1.5 to 3.0. In the case of a shape factor equal to 2, the Weibull distribution is called the Rayleigh distribution. The Rayleigh probability distribution function can be described by Equation (3) [20]:

$$f(v) = \left(\frac{2v}{C^2}\right) \exp\left(-\left(\frac{v}{C}\right)^2\right) \quad (3)$$

The standard deviation σ , which measures how far the wind speed is from the mean speed $\bar{v}$ of the Weibull distribution, and the mean speed can be defined by Equations (4) and (5), respectively [19, 21]:

$$\sigma = \sqrt{\frac{1}{N-1} \sum_{i=1}^N (v_i - \bar{v})^2} \quad (4)$$

and

$$\bar{v} = \frac{1}{N} \sum_{i=1}^N v_i \quad (5)$$

where N is the number of recorded wind speeds.

There are numerous methods for estimating Weibull parameters. In this research, we will present the most common and accurate methods as follows.

1.1.1 Empirical trends of the shape-factor method

Justus *et al.* have determined a general trend of Weibull shape factor k and the mean wind speed through an empirical approach. This empirical approach relies on estimating k from the variability of the wind and the average wind speed as defined in Equation (6) [19, 22]:

$$\text{The shape parameter is defined by : } k = \begin{cases} 1.05v^{0.5} & \text{if } v < 3 \\ 0.94v^{0.5} & \text{if } 3 < v < 4 \\ 0.83v^{0.5} & \text{if } v > 4 \end{cases} \quad (6)$$

and the scale parameter C for all methods (empirical, power density, standard deviation methods) is defined in by Equation (7) [23]:

$$C = \frac{\bar{v}}{\Gamma(1 + \frac{1}{k})} \quad (7)$$

1.1.2 Standard deviation method

This method relies on the availability of the average wind speed and the standard deviation values accordingly the shape factor can be approximately estimated using the following Equation (8) [19, 24]:

$$k = \left(\frac{\sigma}{\bar{v}}\right)^{-1.086} \quad (8)$$

The mean wind speed and standard deviation defined by Equations (9) and (10):

$$\bar{v} = \frac{1}{N} \sum_{i=1}^N v_i \quad (9)$$

and

$$\sigma = \sqrt{\frac{1}{N-1} \sum_{i=1}^N (v_i - \bar{v})^2} \quad (10)$$

and the scale factor can be estimated using Equation (7).

1.1.3 Power density method (PDM)

One of the simplest methods for calculating the Weibull parameters is the PDM. The energy factor E_{pf} can be calculated as the ratio between the mean of the sum of the cubes of all wind speeds and the cube of the mean wind speed as per Equation (12). The simplicity of the equation makes it very useful for initial calculations, but it has also been observed that the fit is not as good as that in other methods when there is low wind speed and high turbulence, which make the Weibull PDM not a smooth curve. Nevertheless, it was chosen for this study as a reference formulation in order to explore the possible differences with more advanced methods.

The shape factor k value is estimated using Equations (11) and (12) [19, 25]:

$$k = 1 + \frac{3.69}{E_{pf}^2} \quad (11)$$

where

$$E_{pf} = \frac{\bar{v}^3}{3} \quad (12)$$

1.1.4 WAsP method

This method has two main conditions:

- The equality of the average power density for both the fitted Weibull distribution and the observed distribution.
- The proportion of values above the mean observed wind speed is the same for the adjusted Weibull distribution and the observed distribution.

The scale parameter defined by first aspect is given by Equation (13) [19, 26]:

$$C = \left[\frac{\sum_{i=1}^N v_i^3}{N\Gamma(1 + \frac{3}{k})} \right]^{1/3} \quad (13)$$

In view of the Weibull distribution function, a proportion of the wind speeds X that exceed the average observed wind speed can be articulated by Equation (14) [26] [10]:

$$X = 1 - F(v_m) = \exp \left[- \left(\frac{v_m}{C} \right)^k \right] \quad (14)$$

The following above such equation is written:

$$-\ln(X) = \left(\frac{v_m}{C} \right)^k \quad (15)$$

Using a combination of Equations (13) and (15), an expression with one unknown (parameter k) is derived by [26]:

$$-\ln(X) = \left[\frac{v_m}{\left[\frac{\sum_{i=1}^N v_i^3}{N\Gamma(1 + \frac{3}{k})} \right]^{1/3}} \right]^k \quad (16)$$

The calculation of X is according to Equation (14), the Weibull shape factor k is estimated iteratively using the Brent method and the Weibull scale factor C is calculated using Equation (13).

1.2 Proper verification and comparison between the different methods

The root mean square error (RMSE) based on the measured data has used to quantify the accuracy of the different methods used to estimate the Weibull parameters as per Equation (17) [27]:

$$RMS \text{ error} = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - x_i)^2} \quad (17)$$

where y_i is the measured value and x_i is the Weibull-parameter-based calculated value.

One year of wind data with 10-minute recording intervals (Wind Atlas of Egypt) has been typically used for estimating Weibull parameters using various estimation methods and direct comparison between them and actual measured data. The wind-measurement campaign was done by the Egyptian New and Renewable Energy Authority (NREA) within the completed preparation of the Egyptian wind atlas and the typical duration of the measurement was 1 year starting from 1 June 2004 to 31 May 2005 with a 99.8% recovery rate [1].

The estimation of Weibull parameters and the RMSE using all methods and measured data from the Elkharga mast is carefully presented in Table 2. It is clear from the results in the table that the WAsP and empirical methods have the lowest RMSE percentage for the site.

Table 2: Estimated scale and shape factors at 25-m height in Elkharga area

	Shape factor k	Scale factor C	RMSE (%)
Empirical method	2.15	7.58	1.23274
Standard deviation method	1.39	7.36	2.98285
Power density method	1.05	6.87	3.90158
WAsP method	2.57	7.4	0.8137

The direct comparison between the best methods (empirical and WAsP methods) and actual measured data is typically presented in Fig. 2. In this specific figure, the WAsP method precisely shows an accurate fitting with the measured frequency distribution at higher wind speeds (>8 m/s) but higher than the actual wind speed at lower wind speeds. However, the empirical method shows lower frequency distribution than the actual measured values at higher wind speeds (>8 m/s) and higher than the actual values at lower wind speeds. These, of course, lead to higher uncertainties in representing the actual site measurements.

Based on the above analysis, results and discussions, it is recommended to adopt the WAsP method for the Elkharga site in this research paper.

1.3 Extrapolations of Weibull parameters from met-mast height to the wind-turbine hub height

Generally, wind speeds are measured using met masts equipped with anemometers fixed at several heights. However, the wind energy captured by the wind turbine is calculated based on the wind speed at the turbine hub height because wind speed varies with the height above ground level. The vertical extrapolation of the wind speed at different heights is formulated by the power-law Equation (18) [24]:

$$v_z = v_0 \left(\frac{h_z}{z_0} \right)^\alpha \quad (18)$$

In Equation (18), v_0 is the measured wind speed at reference height z_0 , v is the wind speed that needs to be calculated at height h_z and the power-law exponent α is calculated using Equation (19) [24, 28]:

$$\alpha = \frac{\ln(\frac{v_2}{v_1})}{\ln(\frac{z_2}{z_1})} \quad (19)$$

where v_1 is the measured average wind speed at the height z_2 and v_2 is the measured wind speed at the height z_1 .

The extrapolation of the Weibull parameters is obtained by [29]:

$$k_z = \frac{k_0}{1 - 0.00881 \ln(\frac{z}{z_0})} \quad (20)$$

$$C_z = C_0 \left(\frac{z}{z_0} \right)^\alpha \quad (21)$$

According to the measured data, the calculated power-law exponent of Elkharga is 0.14 and the shear profile of the site is shown in Fig. 3. Accordingly, the extrapolated average wind speed at 120 m (turbine hub height) is 8.63 m/s.

1.4 Comprehensive assessment of wind characteristics at the study area

The comprehensive assessment of the wind characteristics is essential to select the suitable wind turbine for the site according to the IEC in Table 1 by estimating the annual average wind speed and the turbulence intensity at a wind speed of 15 m/s. In addition, the calculated wind rise illustrated in Fig. 6 shows the

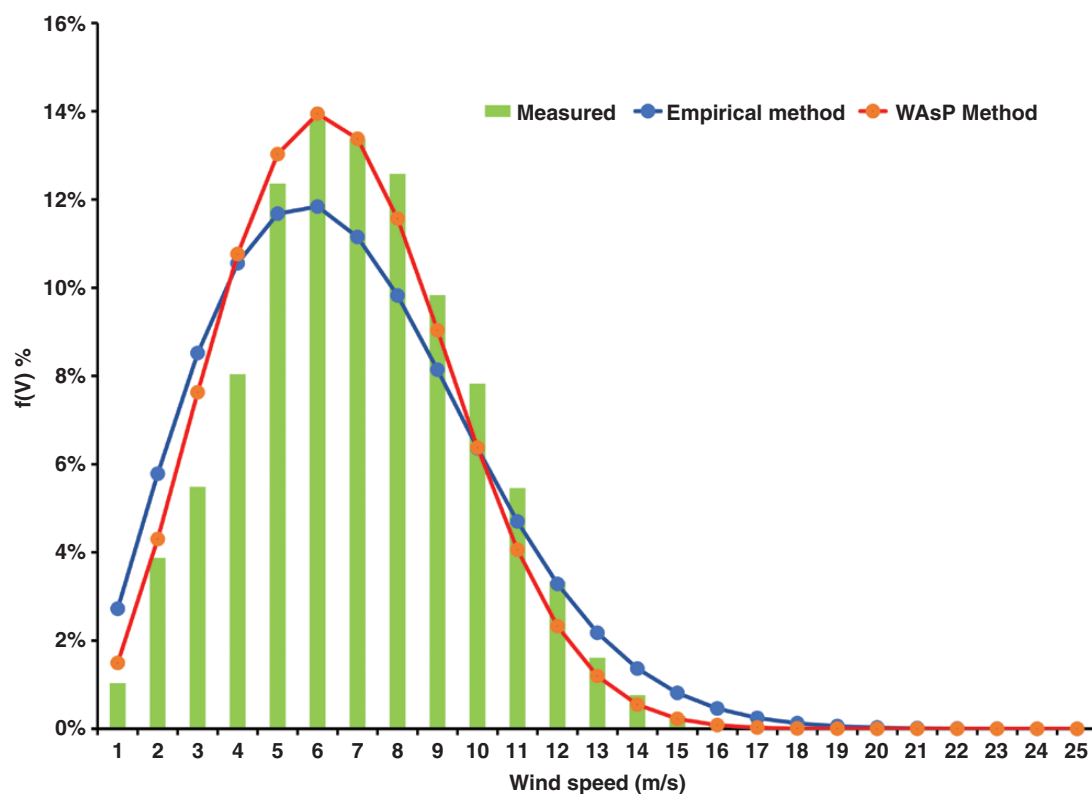


Fig. 2: Comparison between measured and estimated wind-frequency distributions using empirical and WASP methods.

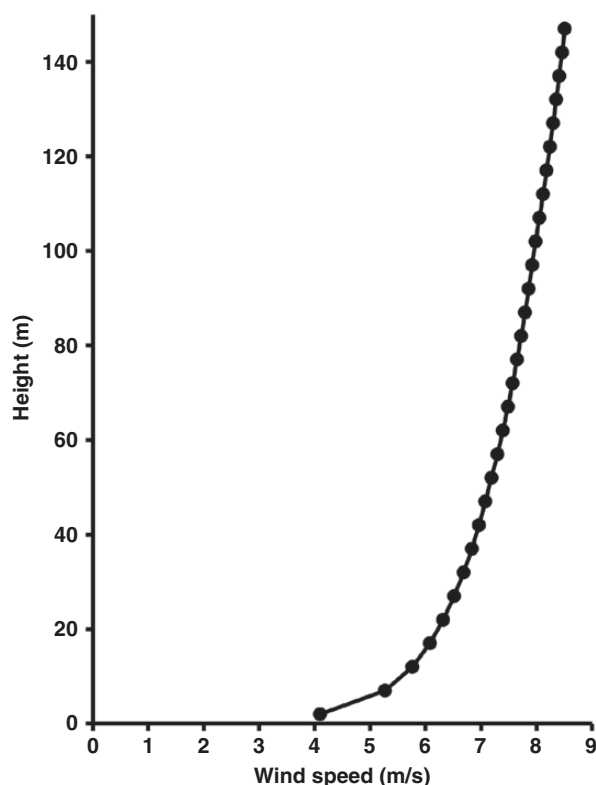


Fig. 3: Wind-speed shear profile.

dominant wind direction, which was typically used for designing the turbines layout within the wind farm. The wind measurement has been obtained from the raw data attached to the Egyptian wind atlas done by Risø National Laboratory in cooperation with

NREA. The site met mast was used to measure wind for 1 year at two height levels of 10 and 25 m, with an interval duration of 10 minutes and recovery rate of 99.8% [1].

1.5 Annual, monthly and daily average wind speeds

Wind speed is a crucial parameter to properly evaluate the wind potential for electricity production from wind turbines. The assessment of wind-speed variability such as the daily, monthly, seasonal and annual variations is extremely important to identify the expected amount of energy that could be produced over a certain period. The possible wind-turbine shutdown periods when the turbine will not be in operation, due to either extremely high or low wind speed, are usually >25 or <3 m/s. The measured wind data have been imported to windPRO [30] as the met-mast object and cleaned from any incorrect data; Fig. 4 illustrates the monthly mean wind speeds of Elkharga at 25 and 10 m above ground level and Fig. 5 shows the daily average wind speed of the Elkharga met mast at 10 and 25 m.

It is shown precisely that the windiest month is February with a mean wind speed of 7.67 m/s at 25 m. However, the lowest wind speed is measured in June with a mean wind speed of 4.72 m/s at the same height. The IEC standard categorizes wind turbines based on the wind speed at turbine hub height and it is assumed to use an intended wind turbine of 120-m hub height; accordingly, the wind speed at 120 m is 8.63 m/s as per Fig. 3 and Equation (18), which means that the site is categorized as Class II. The daily average wind speed in Fig. 5 shows what the peak generation will be from 03:00 pm until 06:00 pm.

1.6 Wind direction

In order to have a proper wind-farm micro-siting avoiding wake losses, it is necessary to know the dominant wind direction. The wind turbines should be aligned in rows and columns with ad-

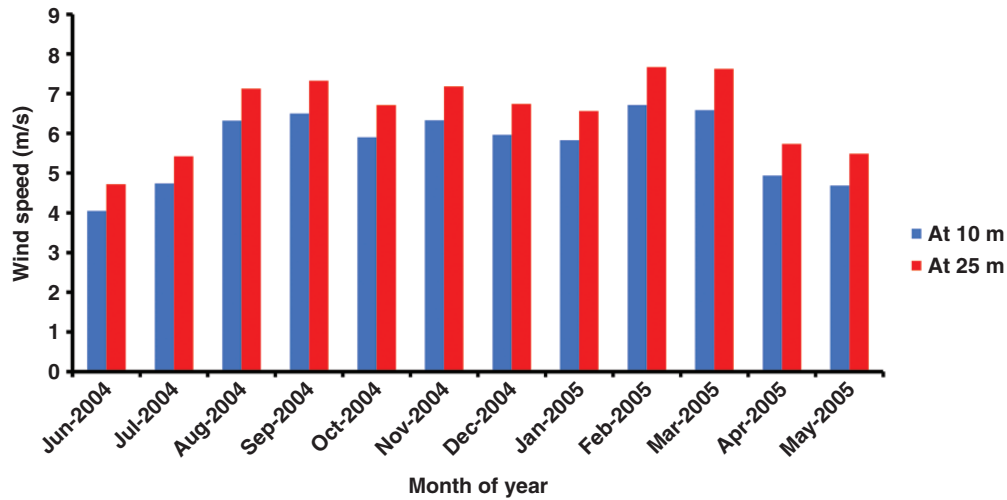


Fig. 4: Monthly average wind speed of the Elkharga met mast at 10 and 25 m (above ground level).

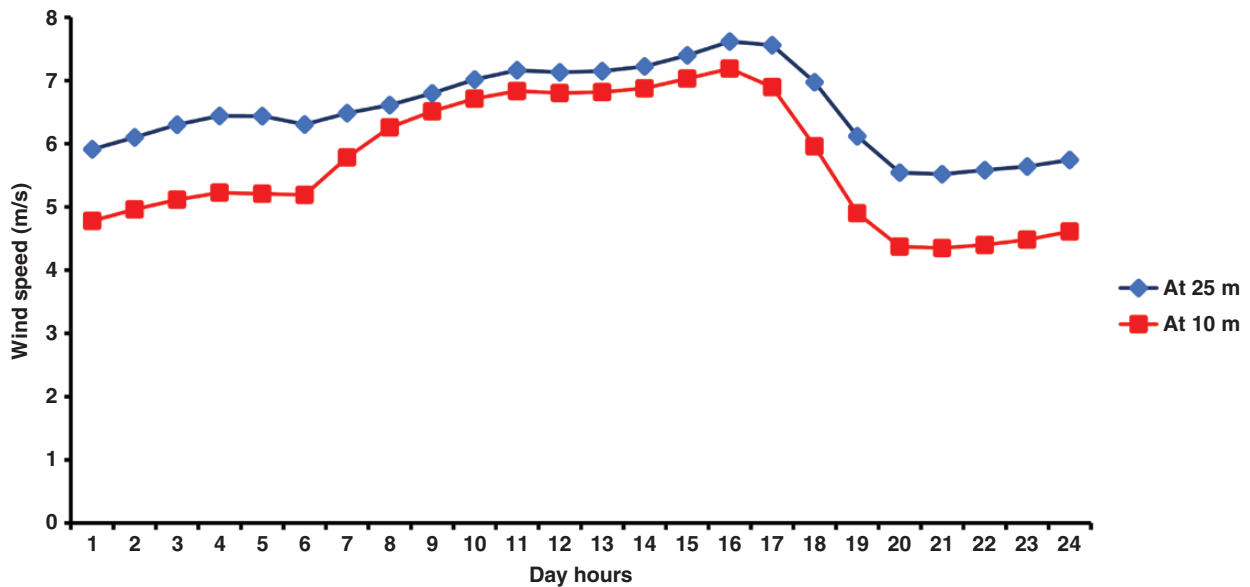


Fig. 5: Daily average wind speed of the Elkharga met mast at 10 and 25 m (above ground level).

equate distancing. The rows must be perpendicular to the dominant wind direction. The wind direction is measured for each wind sector and known as wind rose, as shown in Fig. 6.

1.7 Turbulence intensity

The turbulence intensity is defined as [15, 25]:

$$TI = \frac{\sigma}{v_{avg}} \quad (22)$$

where σ is the standard deviation as defined in Equation (9) and v_{avg} is the annual average wind speed. The turbulence intensity is calculated at the WTG hub height (120 m). The mean turbulence intensity (TI_{15}) at 120 m is 0.053, which typically categorizes the site as Class II B according to the IEC standard in Table 1. Fig. 7 shows the measured turbulence intensity and the type A and B of the IEC 61400-1 standard at different wind speeds extrapolated from windPRO using the measured data.

1.8 Terrain complexity

The land topography has been assessed through an open-access on-line satellite data from the Shuttle Radar Topography Mission. The

Google Earth maps shown in Fig. 8 indicate that the site is of flat terrain, but there are some small hills, sand dunes and slopes that should be considered during the micro-siting of the wind turbines. The advantage of a flat terrain is that it decreases the uncertainties of horizontal extrapolation of the wind measurement across the study area.

1.9 Wind-turbine selection

The minimum criteria for selecting the wind turbine have been precisely defined by IEC 61400 in order to ensure the ideal operation of the wind turbine during the lifetime of the wind farm. There are many turbine manufacturers that provide several types, versions and capacities of wind turbines that can comply with the IEC requirement for the wind site. According to the site wind characteristics and the specific criteria in Table 1, the site is categorized under Class II B. Therefore, three wind-turbine models have been selected, namely V110, V150 and V162, with standard specifications explained in Table 3 and power curves shown in Fig. 9.

1.10 The proposed wind-farm design approach

The proper designing of the wind farm is extremely important to develop wind projects and has a considerable impact on the

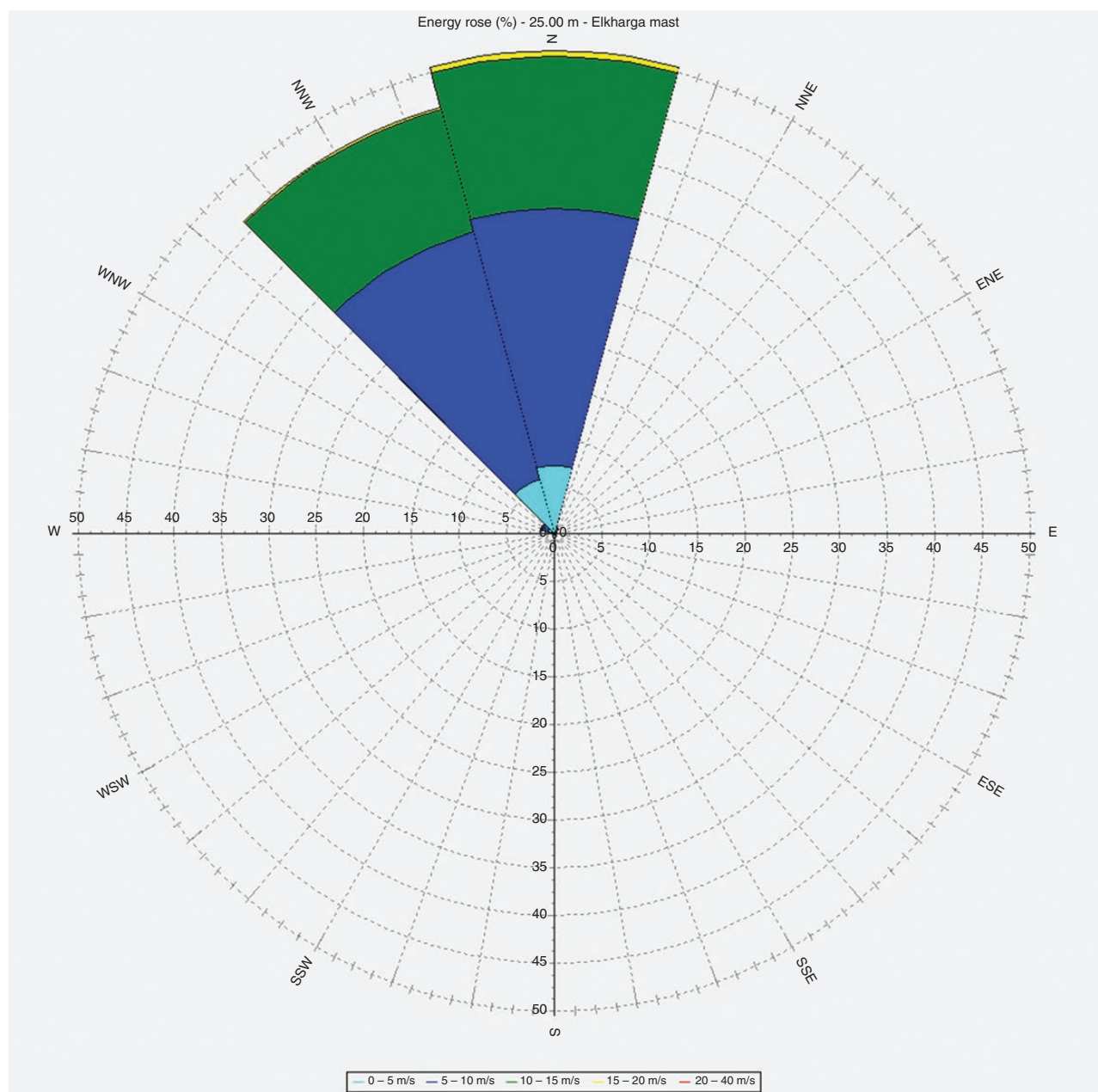


Fig. 6: Elkharga mast measured wind rose.

operational stage. The optimization of placing wind turbines has an effective role in increasing the annual production of the wind farm leading to an increase in financial revenues. The layout optimization was performed using the windPRO optimization module and the key steps of the design of the layout can be summarized as follows:

- identifying the border of the study area including all site restrictions such as environmental zones, buffers from public roads and transmission lines as shown in Fig. 1;
- specifying the maximum installed capacity for the wind farm (50 MW), the considered number of turbines is 25 of the V110 type and 9 turbines of the V150 and V162 types, the distancing between turbines and the orientation perpendicular to the dominant wind direction. The distancing between turbines for a wind site that has a dominant wind

direction usually has the shape of an ellipse based on the two alternatives:

Alternative 1: the ellipse minor axis is $3 \times RD$, the major axis is $10 \times RD$ and the angle is 345° ;

Alternative 2: the ellipse minor axis is $2 \times RD$, the major axis is $10 \times RD$ and the angle is 345° ;

- preparing a wind-resource map with a resolution of 50×50 m over the study area to allow the optimizer module to select the best turbine locations.

All these issues of our design approach have been imbedded in the windPRO optimization model for the proposed wind farm at Elkharga Oasis in Egypt.

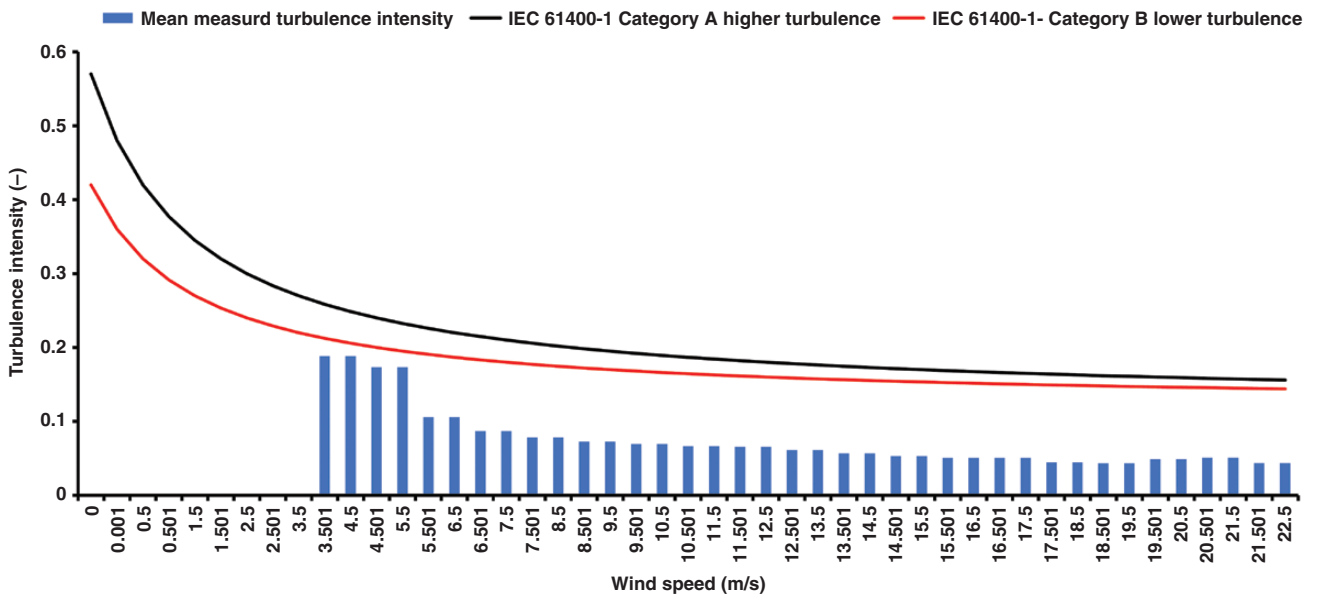


Fig. 7: Measured turbulence intensity at 120 m and the IEC standard values.

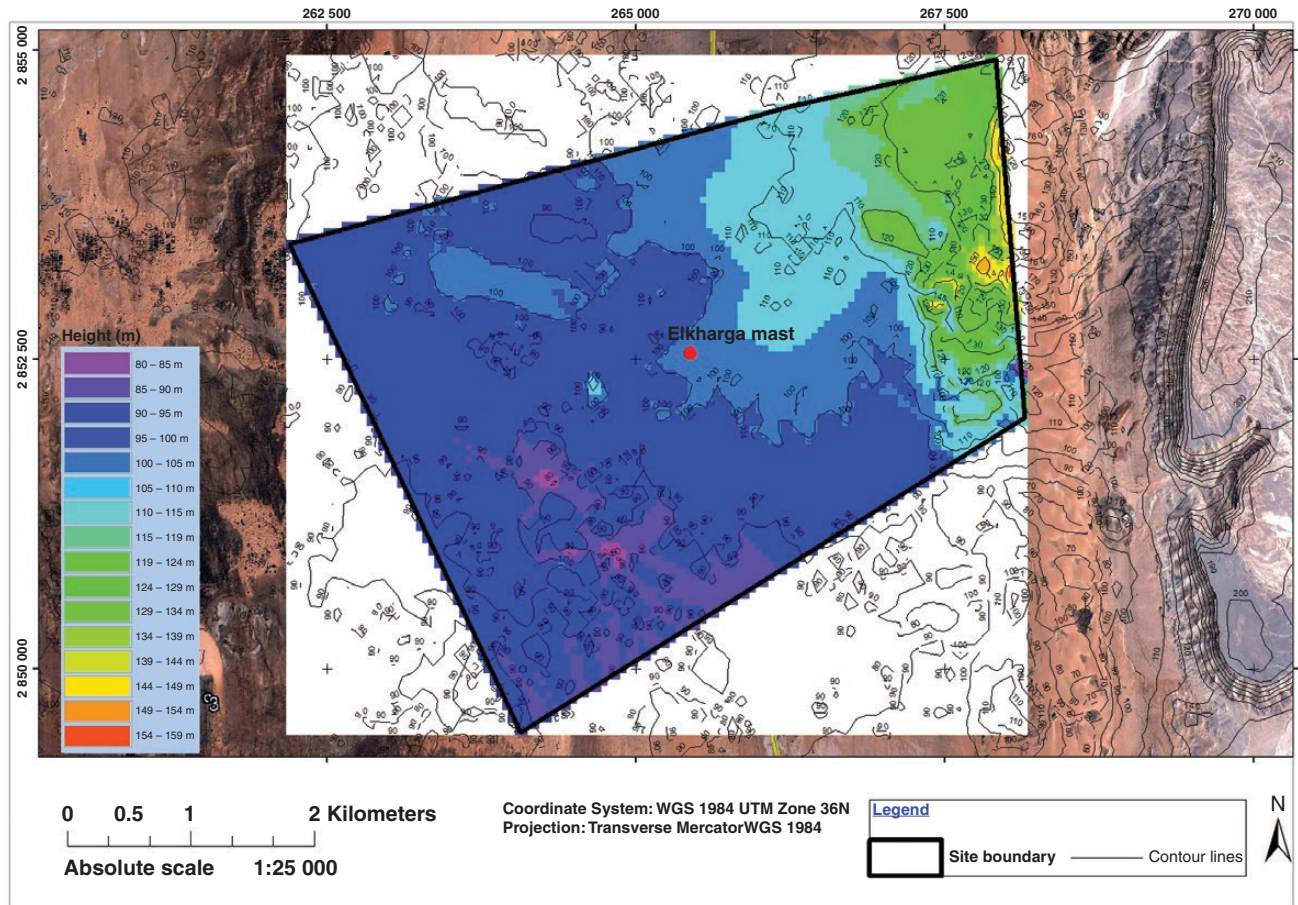


Fig. 8: Land-elevation levels above ground level of the study area [1].

Details of results and discussions are given in the following sections.

1.11 Calculation methods of the expected AEP of the proposed Elkharga wind farm

Based on the above design approach, the estimation of the expected AEP from the proposed Elkharga wind farm will be introduced.

The power output from any wind turbine is a function of the wind speed, which comes from the turbine power curves provided by the wind-turbine manufacturers. The standard wind-turbine power curve provides the rated power (P_r), cut-in (v_{ci}), rated (v_r) and cut-out (v_{co}) wind speeds. The wind turbine starts generating power when the speed exceeds the cut-in speed (v_{ci}) until reaching the rated speed (v_r) at which generation of the rated power (P_r)

remains constant up to the cut-out (v_{co}) speed, when the turbine shuts down.

1.12 The annual production of the wind farm can be estimated using two methods

1.12.1 Method 1

The annual production of an individual wind turbine is the product of the Weibull probability density function described in Equation (1), the turbine power curve and the number of hours per year. The whole wind-farm production is the product of the individual turbine production and the number of turbines at the wind farm as per Equation (23). The Weibull distribution parameters should be calculated using the WAsP method and extrapolated to the turbine hub height as described in Fig. 2 [31]:

$$E = N \times \sum_{i=v_{ci}}^{i=v_{co}} P(v_i) \times f(v_i) \times 8760 \quad (23)$$

where E represents the AEP (kWh), N represents the number of turbines at the wind farm, $P(v_i)$ represents the turbine power output (kW), v_i represents the wind speed (m/s) and $f(v)$ represents the Weibull probability density function.

In complex terrain, the wind speed is not similar over the whole site and so this method will be inaccurate because it assumes similar wind characteristics for the whole site, and it needs to assume a certain value of wake losses in the case of using multiple turbines that cause wake interaction.

1.12.2 Method 2

The annual yield is accurately calculated using the windPRO software, which will be the appropriate way in the case of estimating the yield of a 50-MW wind farm that typically consists of multiple turbines. The windPRO software creates a general wind statistic for the site using the measured wind data and extrapolates it horizontally at each turbine position and vertically at turbine hub heights using the land orography and roughness maps. In

addition, the windPRO software typically has an advanced model to calculate the internal and external wake losses.

Direct comparison between the two prevalent methods will be according to the AEP and capacity factor C_f of each prevalent method and the C_f is calculated using Equation (24) [32]:

$$C_f = \frac{E_0}{P_r \times 8760} \times 100\% \quad (24)$$

where E_0 is the estimated AEP (kWh) and P_r is the rated power output.

1.13 Economic investigation of Elkharga wind turbines based on the LCOE

The LCOE is extensively used to accurately calculate the cost of electricity from various energy sources. The economic analysis is concluded by estimating the LCOE for each turbine type to correctly decide which turbine is the most competitive. The LCOE considers the sum of all expenditures over the project lifetime, discounted to the present time and levelized based on the AEP [2, 33–35]:

$$LCOE = \frac{\sum_{t=1}^n \frac{I_t + M_t + F_t}{(1+r)^t}}{\sum_{t=1}^n \frac{E_t}{(1+r)^t}} \text{ \$/MWh} \quad (25)$$

where LCOE is the average lifetime levelized cost of electricity generation, I_t is the investment cost in year t , M_t is the O&M expenditures in the year t , F_t is the fuel expenditures in the year t , E_t is the electricity generation in the year t in kWh or MWh, r is the discount rate as a percentage and n is the life time of the project in years.

Like most renewable-energy technologies, the investment cost of wind energy is high. The capital expenditure (CapEx) is the biggest portion of the life-cycle costs of a wind farm. The prevailing trend of steadily decreasing CapEx in recent years is due to a number of key factors such as turbine upsizing and material labour costs. The wind-farm CapEx consists of the wind-turbine cost, which typically represents the most costly item, and the electrical and civil balance of the plant. The actual wind-turbine cost is not fixed and is negotiable based on each individual project and most turbine suppliers and independent developers do not reveal it. According to publicly available data sources of wind-turbine pricing reports, it is assumed that the capital cost of the V110–2 MW is \$1013/kW, assuming additional expected reduction for the V150 and V162 by 15% and 20%, respectively, due to expected savings from scaling up the wind turbine [33].

Table 3: Specifications of selected wind turbines

Turbine name	Capacity (MW)	Hub height (m)	Rotor diameter (m)
V110	2	120	110
V150	5.6	120	150
V162	5.6	120	162

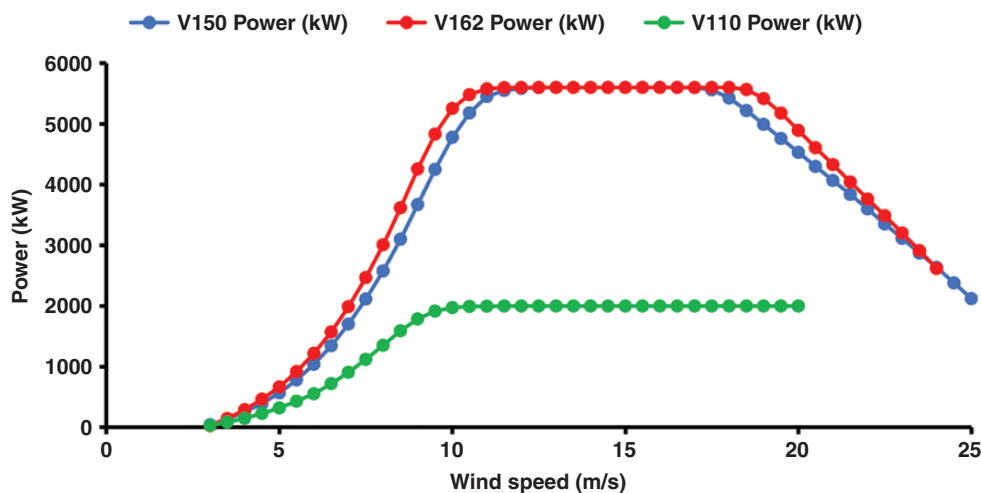
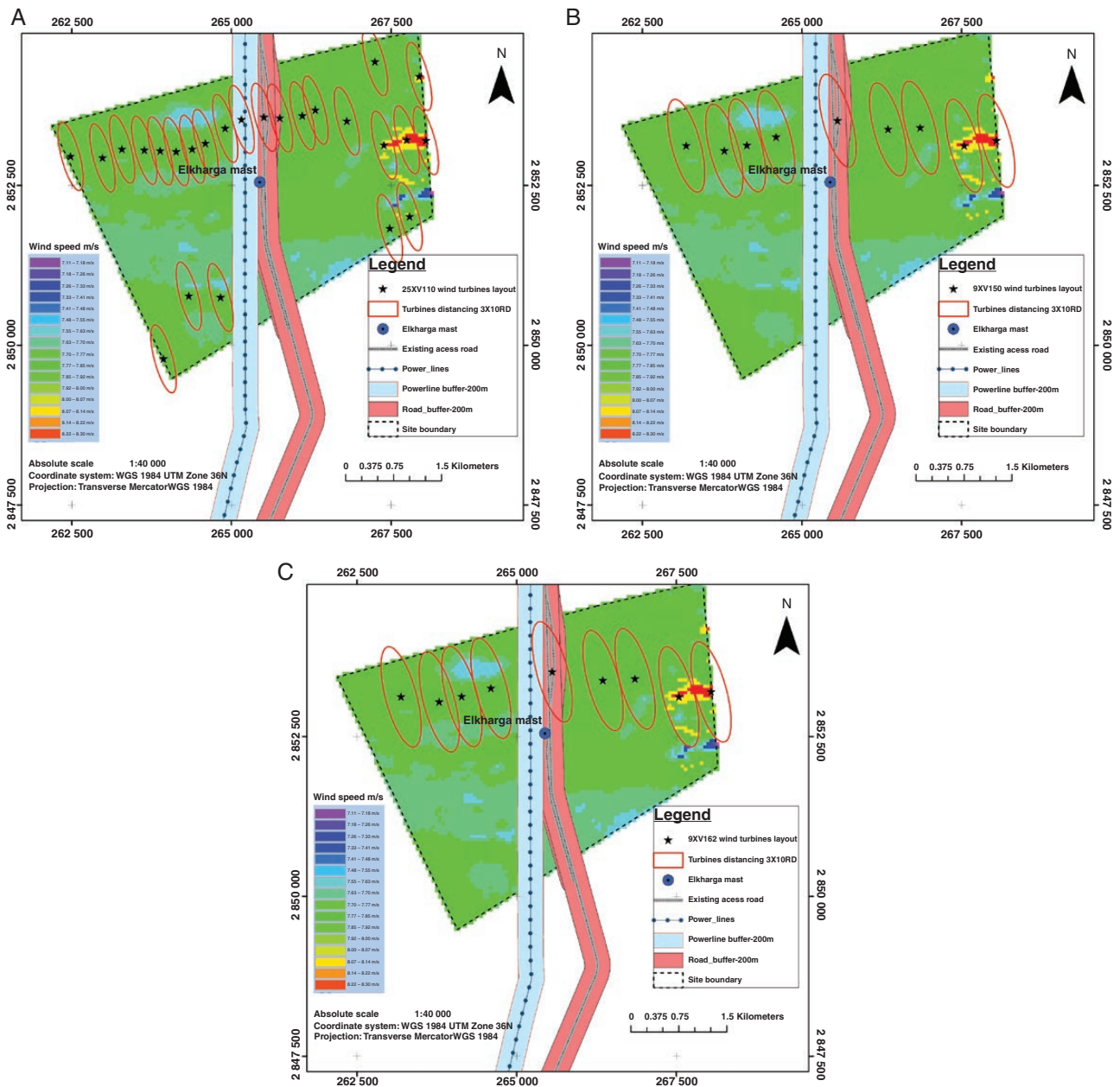


Fig. 9: The power curves of the shortlisted wind turbines.

Table 4: Comparison between the two yield-calculation methods

Turbine model	Annual yield calculation using Method 1		Annual yield calculation using Method 2	
	V110 (Configuration 1)	V110 (Configuration 2)	V110 (Configuration 1)	V110 (Configuration 2)
Hub height (m)	120	120	120	120
Turbine capacity (MW)	2	50	2	50
Number of turbines	1	25	1	25
Annual production (GWh) excluding additional losses	10.635	265.883	10.429	247.5017
Capacity factor %	61	61	59.5	56.5
Delta percentage between Methods 1 and 2 (%)	+1.9	+6.9	-1.98	-7.4


Fig. 10: Optimized wind turbines layout using the Alternative 1 layout (3 × 10 RD). (a) 25 × V110-2 MW wind turbines layout; (b) 9 × V150-5.6 MW wind turbines layout; (c) 9 × V162-5.6 MW wind turbines layout.

2 Results and discussion

In Table 4, the AEP and capacity factor from the V110 wind turbine are calculated according to Methods 1 and 2 using two different turbine configurations. The first configuration assumes that the wind farm consists of one turbine located at the mast position while the second configuration assumes 25 turbines for the Elkharga wind farm. The results show that the two methods are identical in the case of using one turbine, but in the case of multiple numbers of turbines, there are big gaps between them. There are typically two main reasons for these gaps; the first possible reason is that Method 1 assumes similar wind conditions over the site area, but in reality, the measured wind statistics represent only the mast position, which needs to be extrapolated horizontally to turbine positions in order to represent the whole site. The second crucial reason is that Method 1 has considered no

wake effect and interaction between turbines, but Method 2 has calculated the internal wake losses using windPRO wake models and deducted them from the gross production. For the sake of economic analysis, this research adopted the windPRO software package.

According to the estimated delta percentage between Methods 1 and 2, it could be possible to use Method 1, which is simple and quick to calculate the expected AEP from any other wind site but considering a certain amount of uncertainty.

In Fig. 10, the windPRO optimizer has properly distributed the wind turbines layout for each turbine model (V110–2 MW, V150–5.6 MW and V162–5.6 MW) according to the conditions of Alternative 1 and Fig. 11 according to Alternative 2 conditions. The wind speed across the study area varies from 7.8 to 8.18 m/s at 120-m height including a few high-wind spots of 8.63 m/s at

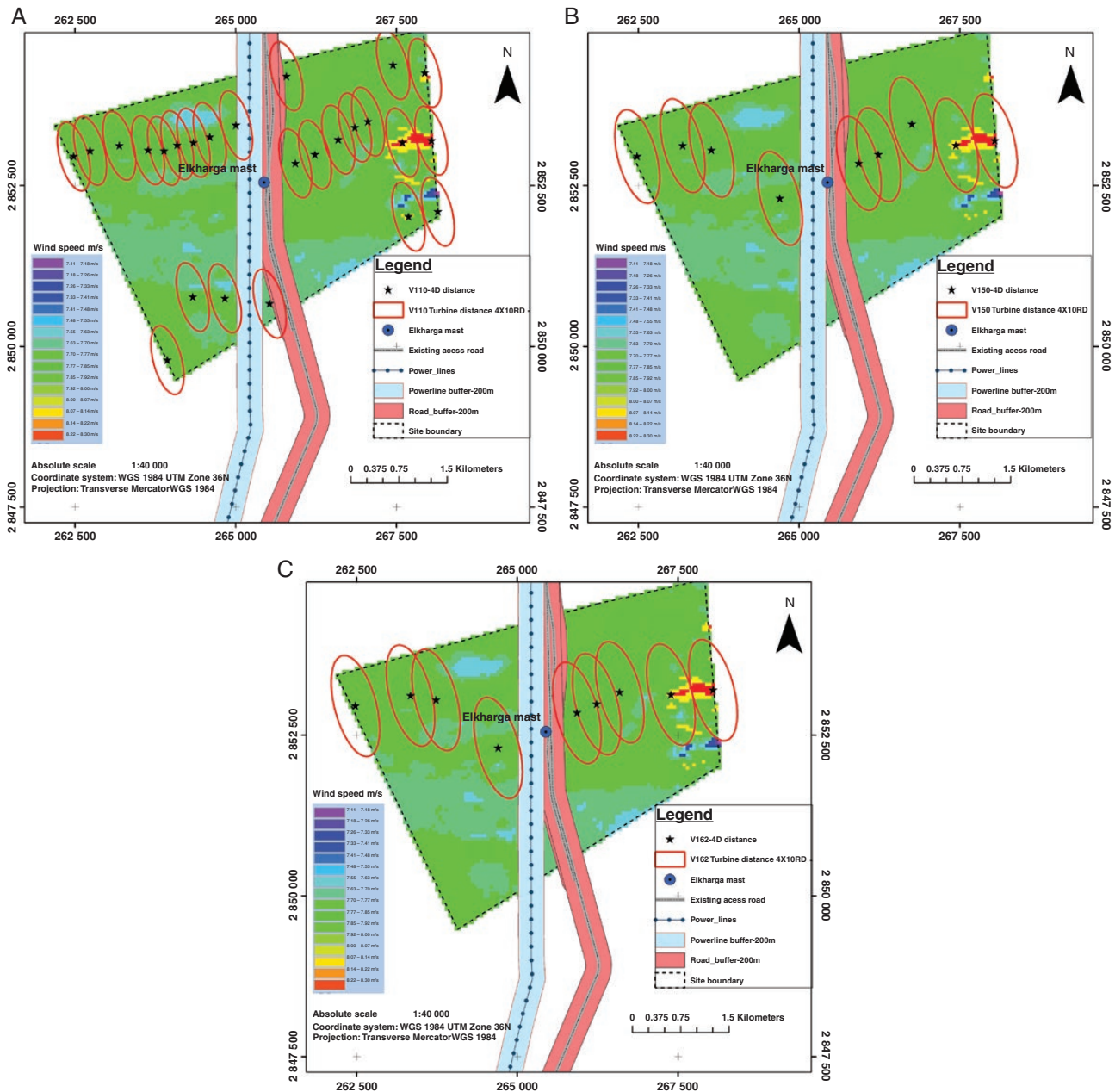


Fig. 11: Optimized wind turbines layout using the Alternative 2 layout (4 × 10 RD). (a) 25 × V110–2 MW wind turbines layout; (b) 9 × V150–5.6 MW wind turbines layout; (c) 9 × V162–5.6 MW wind turbines layout.

Table 5: Annual energy production and capacity factor for each layout configuration

Layout configuration	V110–2 MW		V150–5.6 MW		V162–5.6 MW	
	3 × 10 RD	4 × 10 RD	3 × 10 RD	4 × 10 RD	3 × 10 RD	4 × 10 RD
Annual gross energy (GWh)	254.977	255.104	215.767	215.068	233.756	233.052
Internal wake losses (%)	2.8	2	1.1	0.4	1.6	0.4
Net wake energy (GWh) excluding other losses	247.942	250.102	213.403	214.229	230.056	232.170
Capacity factor (C _p) (%)	56.6	57.1	48.3	48.5	52.1	52.6
Overall increase in yield due to increasing from 3 to 4 RD (%)	+0.87		+0.391		+0.918	

Table 6: The technical inputs and assumptions to estimate the LCOE for each layout of a 50-MW wind farm

Layout configuration	V110–2 MW		V150–5.6 MW		V162–5.6 MW	
	3 × 10 RD layout	4 × 10 RD layout	3 × 10 RD layout	4 × 10 RD layout	3 × 10 RD layout	4 × 10 RD layout
Total installed capacity (MW)	50	50	50.4	50.4	50.4	50.4
CapEx (USD/kW)	1013	1013	861	861	810	810
Availability loss factor (%)	97	97	97	97	97	97
Other losses factors (electrical losses, grid availability, environmental, etc.) (%)	94	94	94	94	94	94
AEP (kWh)	233 065 480.00	235 095 880.00	200 598 820.00	201 375 260.00	216 252 640.00	218 239 800.00
Lifetime of the project (N-years)	20	20	20	20	20	20
Equity/debt	20/80%	20/80%	20/80%	20/80%	20/80%	20/80%
Equity cost (%)	15	15	15	15	15	15
Debt interest (%)	6.3	6.3	6.3	6.3	6.3	6.3
Discount factor (%)	8	8	8	8	8	8
O&M cost	4.2% of CapEx	4.2% of CapEx	3% of CapEx	3.5% of CapEx	3.5% of CapEx	3.5% of CapEx
LCOE \$/MWh	34.2861	33.99002	32.4701	32.3449	28.4104	28.1517

120 m located close to the eastern borders. The annual production is calculated using windPRO and properly presented in Table 5. The wind turbines layout optimization process seeks an arrangement of the turbines that leads to utilizing the highest wind potential and concurrently maintaining the wake interactions at the most elementary level. The annual production is calculated using windPRO and presented in Table 5, and the overall increase in AEP due to proportionately increasing the distance between the turbines in the same row from 3 × RD to 4 × RD ranges from 0.31% for the V150, 0.87% for the V110 and 0.918% for the V162. It is essential to do cost-benefit analysis between increasing the balance-of-plant costs such as road, cables and electrical cable losses against the gain from increasing the annual production during the lifetime of the wind farm.

The study area has the advantage of flat terrain, but for other sites characterized by complex terrain, the possibility of yield improvement benefits through increasing wind-turbine distances could be limited and it may need a detailed load assessment on wind-turbine components.

In Table 6, the capital and operation expenditures and therefore the financial cost have been reasonably assumed to accurately estimate the levelized cost of electricity per turbine model and layout. For the larger turbine models like the V150 and V162, lower capital expenditures \$/MW and the O&M costs lower than those of the smaller turbine V110 have been assumed, because scaling up wind turbines leads to more competitiveness in relation to capital cost and the wind farm will have a lower number of turbines that are definitely simpler to operate and properly manage [33]. The LCOE of the V162 turbine has the lowest value, ranging from 28.1517 to 28.4104 \$/MWh, depending on the turbine

distancing. However, the V110 had the highest LCOE range from 33.99 to 34.2861 \$/MWh.

Based on these results and discussions, the best wind turbines at the Elkharga site can be ranked in decent order as V162, V150 and V110.

3 Conclusion

The Egyptian western desert has enormous wind-energy potential, especially close to the Elkharga Oasis, where the wind speed reaches 8.63 m/s at 120 m. This potential can be properly utilized to generate electricity and undoubtedly contribute well to electricity demand in this remote area.

The proposal of a new wind-farm project in that area can improve tremendously the social, environmental and economic conditions for the local community in the region. In addition, the new wind farm will excessively reduce the reliance on generating electricity from conventional power plants that typically use oil and gas, which have price fluctuation over time.

The ultimate conclusion of this research has clearly proved the technical and economic viability of properly establishing a wind-farm project that may attract the attention of public and private developers.

The wind potential is assessed using the statistical method of the Weibull probability density function. The methods of estimating the Weibull distribution parameters are subject to the wind characteristics of the site, so it is essential to assess each method and verify it with actual measured values before calculating the expected AEP. For the Elkharga site, the WASP method has shown the lowest RMSE.

Based on the IEC standard and the emphasis on the levelized cost of electricity, the study of the 50-MW wind farm at Elkharga shows the possibility to generate electricity from the wind farm at a levelized cost of ~\$28.15/MWh.

In comparison to the current tariff from utility companies in Egypt (\$66.87/MWh at the medium voltage level), this research concludes that the energy cost of the proposed wind farm at Elkharga is much cheaper by >50% than the utility price.

Finally, it is essential to consult the local communities and investigate the social impact of large-scale wind-turbine deployment in the region.

Conflict of interest statement

None declared.

References

- [1] Mortensen NG, Hansen JC, Badger J, et al. Wind atlas for Egypt: measurements, micro-and mesoscale modelling. In: *European Wind Energy Conference & Exhibition*, Vol. 27, Athens, Greece, 27 February–2 March 2006, 656.
- [2] Shields M, Beiter P, Nunemaker J, et al. Impacts of turbine and plant upsizing on the levelized cost of energy for offshore wind. *Applied Energy*, 2021, 298:117189.
- [3] Alemzero D, Acheampong T, Huaping S. Prospects of wind energy deployment in Africa: technical and economic analysis. *Renewable Energy*, 2021, 179:652–666.
- [4] Mohamadi H, Saeedi A, Firoozi Z, et al. Assessment of wind energy potential and economic evaluation of four wind turbine models for the east of Iran. *Heliyon*, 2021, 7:e07234.
- [5] Dabar OA, Awaleh MO, Kirk-Davidoff D, et al. Wind resource assessment and economic analysis for electricity generation in three locations of the Republic of Djibouti. *Energy*, 2019, 185:884–894.
- [6] Himri Y, Merzouk M, Kasbadji Merzouk N, et al. Potential and economic feasibility of wind energy in south West region of Algeria. *Sustainable Energy Technologies and Assessments*, 2020, 38:100643.
- [7] Bilal B, Adjallah KH, Yetilmmezsoy K, et al. Determination of wind potential characteristics and techno-economic feasibility analysis of wind turbines for Northwest Africa. *Energy*, 2021, 218:119558.
- [8] El Khchine Y, Sriti M, Elyamani NEEK. Evaluation of wind energy potential and trends in Morocco. *Heliyon*, 2019, 5:e01830.
- [9] Yilmaz U, Balo F, Sua LS. Simulation framework for wind energy attributes with WAsP. *Procedia Computer Science*, 2019, 158:458–465.
- [10] Katinas V, Gecevicius G, Marciukaitis M. An investigation of wind power density distribution at location with low and high wind speeds using statistical model. *Applied Energy*, 2018, 218:442–451.
- [11] Rogers T, Ashtine M, Koon RK, et al. Onshore wind energy potential for small island developing states: findings and recommendations from Barbados. *Energy for Sustainable Development*, 2019, 52:116–127.
- [12] Elia A, Taylor M, Ó Gallachóir B, et al. Wind turbine cost reduction: a detailed bottom-up analysis of innovation drivers. *Energy Policy*, 2020, 147:111912.
- [13] Katinas V, Sankauskas D, Markevičius A, et al. Investigation of the wind energy characteristics and power generation in Lithuania. *Renewable Energy*, 2014, 66:299–304.
- [14] Hamouda YA. Wind energy in Egypt: economic feasibility for Cairo. *Renewable and Sustainable Energy Reviews*, 2012, 16:3312–3319.
- [15] Katsigiannis YA, Stavrakakis GS, Pharconides C. Effect of wind turbine classes on the electricity production of wind farms in Cyprus island. *Conference Papers in Energy*, 2013, 2013:1750958.
- [16] Leu T-S, Yo J-M, Tsai Y-T, et al. Assessment of IEC 61400-1 normal turbulence model for wind conditions in Taiwan west coast areas. *International Journal of Modern Physics: Conference Series*, 2014, 34:1460382.
- [17] Bilal BO, Ndongo M, Kebe CMF, et al. Feasibility study of wind energy potential for electricity generation in the northwestern coast of Senegal. *Energy Procedia*, 2013, 36:1119–1129.
- [18] Baseer MA, Meyer JP, Alam MM, et al. Wind speed and power characteristics for Jubail industrial city, Saudi Arabia. *Renewable and Sustainable Energy Reviews*, 2015, 52:1193–1204.
- [19] Sriti M. Performance evaluation of wind turbines for energy production in Morocco's coastal regions. *Results in Engineering*, 2021, 10:100215.
- [20] Pishgar-Komleh SH, Keyhani A, Sefeedpari P. Wind speed and power density analysis based on Weibull and Rayleigh distributions (a case study: Firouzkoo county of Iran). *Renewable and Sustainable Energy Reviews*, 2015, 42:313–322.
- [21] Bilir L, İmri M, Devrim Y, et al. An investigation on wind energy potential and small scale wind turbine performance at İncek region—Ankara, Turkey. *Energy Conversion and Management*, 2015, 103:910–923.
- [22] Justus C, Hargraves W, Mikhail A, et al. Methods for estimating wind speed frequency distributions. *Journal of Applied Meteorology*, 1978, 17:350–353.
- [23] Wan C, Wang J, Yang G, et al. Wind farm micro-siting by Gaussian particle swarm optimization with local search strategy. *Renewable Energy*, 2012, 48:276–286.
- [24] Manwell JF, McGowan JG, Rogers AL. *Wind Energy Explained: Theory, Design and Application*. 2nd edn. Chippingham: John Wiley & Sons, 2010.
- [25] Raichle BW, Carson WR. Wind resource assessment of the Southern Appalachian Ridges in the Southeastern United States. *Renewable and Sustainable Energy Reviews*, 2009, 13:1104–1110.
- [26] Solyali D, Altunç M, Tolun S, et al. Wind resource assessment of Northern Cyprus. *Renewable and Sustainable Energy Reviews*, 2016, 55:180–187.
- [27] Baseer MA, Meyer JP, Rehman S, et al. Wind power characteristics of seven data collection sites in Jubail, Saudi Arabia using Weibull parameters. *Renewable Energy*, 2017, 102:35–49.
- [28] Xu C, Hao C, Li L, et al. Evaluation of the power-law wind-speed extrapolation method with atmospheric stability classification methods for flows over different terrain types. *Applied Sciences*, 2018, 8:1429.
- [29] Gökçek M, Bayülken A, Bekdemir Ş. Investigation of wind characteristics and wind energy potential in Kırklareli, Turkey. *Renewable Energy*, 2007, 32:1739–1752.
- [30] EMD International A/S. windPRO. <https://www.emd-international.com/windpro/> (6 January 2022, date last accessed).
- [31] Astanei D, Nemes C, Munteanu F, et al. Annual energy production estimation based on wind speed distribution. In: *International Conference and Exposition on Electrical and Power Engineering (EPE)*, Iasi, Romania, 20–22 October 2016, 862–867.
- [32] Olaofe ZO, Folly KA. Wind energy analysis based on turbine and developed site power curves: a case-study of Darling City. *Renewable Energy*, 2013, 53:306–318.

- [33] Stehly T, Beiter P, Duffy P. 2019 *Cost of Wind Energy Review*. Golden, CO: National Renewable Energy Lab (NREL), 2020.
- [34] Quinn R, Schepers G, Bulder B. A parametric investigation into the effect of low induction rotor (LIR) wind turbines on the levelised cost of electricity for a 1 GW offshore wind farm in a North Sea wind climate. *Energy Procedia*, 2016, 94:164–172.
- [35] Johnston B, Foley A, Doran J, et al. Levelised cost of energy: a challenge for offshore wind. *Renewable Energy*, 2020, 160:876–885.